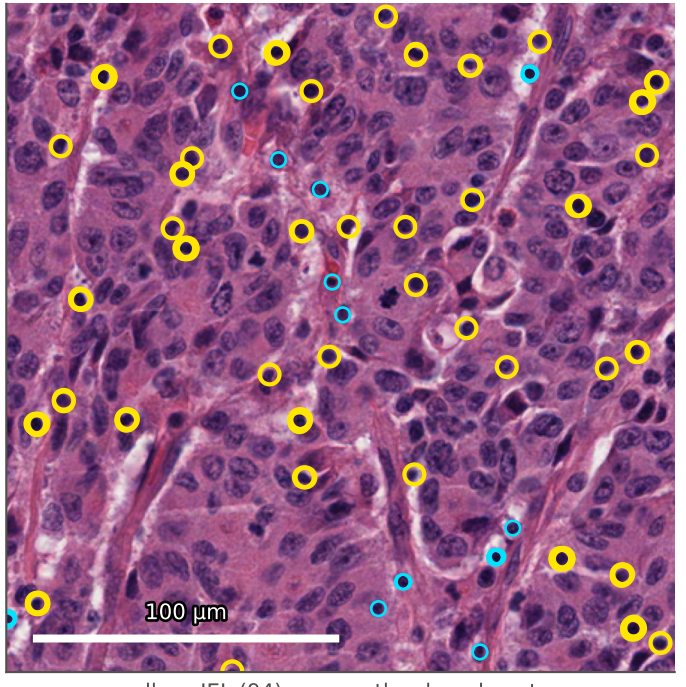
a Established marker

count intraepithelial lymphocytes (IEL)



yellow: IEL (94); cyan: other lymphocytes

b Foundation-model embeddings

pool tile embeddings per patient: mean | max | attention

MSI training patient
1,188 tiles released

MSS training patient
185 tiles released



cyan: tiles in the released dataset (similar tumor size, same site)

Pitfall 1: tile-count shortcut

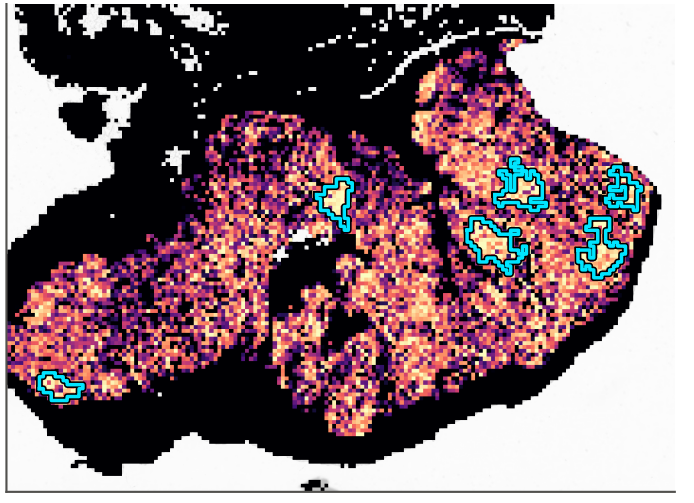
MSS training tiles were discarded to balance classes; max pooling learns tile count.

Pitfall 2: wrong resolution

Tiles documented and reused as 0.5 µm/px are 1.14 µm/px.

c Attention-guided discovery

let attention point, pathologists name



UNI2-h ABMIL attention (bright = high); cyan: regions sent to blinded review by two pathologists

d Which approach stratifies MSI across hospitals?

Identical tumor tiles (≤1,000 per patient; Prov-GigaPath-flash embeddings, KongNet nuclei); all models fit on TCGA training patients only

